AI Interview Chatbot

Project Requirements & Technical Report

Item	Detail
Product	Voice-based, proctored AI mock-interview portal
AI model	Google Gemini 2.5 Flash (free tier)
Stack	Next.js 15, TypeScript, Tailwind CSS, Framer Motion
Version	1.0 (Requirements baseline)
Date	15 October 2026

1. Introduction

The AI Interview Chatbot is a web portal where anyone can practise a realistic job interview. The candidate states the role they are targeting and their years of experience, picks how long the session should be (by number of questions or by time), and is then interviewed by an AI interviewer that speaks its questions aloud and listens to spoken answers. The session runs in a proctored, full-screen environment; leaving the tab or full screen costs 15 marks per violation. At the end the candidate receives a detailed, downloadable performance report.

1.1 Objectives

- Give job seekers a free, realistic, voice-first interview practice tool.
- Tailor questions to the target role and experience level using Gemini.
- Simulate exam-like discipline through proctoring (full screen, no tab switching).
- Provide actionable feedback: per-question scores, strengths, weaknesses and model answers.
- Deliver a premium, modern and user-friendly interface that works on desktop browsers.

1.2 Scope

In scope: anonymous access (no login), interview setup, voice interview, proctoring, AI evaluation, result report with PDF download. **Out of scope (v1):** user accounts, saved history, video recording, recruiter dashboards, mobile-native apps.

2. Target Users

User	Need
Students / freshers	Practise first interviews and build confidence speaking answers aloud.
Experienced professionals	Prepare for senior or role-switch interviews with harder, scenario-based questions.
Career switchers	Understand what is expected for a new role and get targeted feedback.

3. Functional Requirements

3.1 Landing Page

- Hero section explaining the product, a clear 'Start Interview' call-to-action and a 3-step 'How it works'.
- Feature highlights: voice interview, AI feedback, proctored mode.
- Browser compatibility notice (Chrome / Edge recommended for voice).

3.2 Interview Setup (multi-step wizard)

ID	Requirement
FR-1	Candidate enters or selects a **target role** (free text with popular suggestions, e.g. Frontend Developer, Data Analyst, Product Manager).
FR-2	Candidate selects **experience**: Fresher (0), 1-2, 3-5, 6-10, 10+ years.
FR-3	Optional: candidate selects key skills / tech stack and difficulty focus to further tailor questions.

ID	Requirement
FR-4	Candidate chooses **interview mode**: *By Questions* (3, 5, 8, 10, 15) or *By Time* (5, 10, 15, 20, 30 minutes).
FR-5	System check step: microphone permission, camera permission, speaker test and voice-recognition support.
FR-6	Rules screen listing proctoring rules; candidate must accept before the interview starts in full screen.

3.3 Voice Interview

ID	Requirement
FR-7	AI interviewer greets the candidate and asks questions **aloud** (text-to-speech); question text is also shown on screen.
FR-8	Candidate answers **by voice**; a live transcript appears as they speak (speech-to-text).
FR-9	Candidate can stop recording and submit; or re-record before submitting. A 'Skip question' option is available.
FR-10	Questions are generated by Gemini one at a time, adapting to role, experience and previous answers (mix of technical, behavioural and situational).
FR-11	Question mode ends after N questions; Time mode shows a countdown and ends when time runs out (the current answer is still evaluated).
FR-12	Animated AI avatar / voice visualiser indicates speaking, listening and thinking states.
FR-13	Candidate can end the interview early (with confirmation).

3.4 Proctoring

ID	Requirement
FR-14	The interview runs in **full-screen mode**; it cannot start without entering full screen.
FR-15	Each **tab switch / window blur** is a violation: **-15 marks** from the final score.
FR-16	Each **exit from full screen** is a violation: **-15 marks**; a blocking overlay asks the candidate to return to full screen.
FR-17	**Auto-termination** after 3 violations (configurable); the report marks the session as terminated.
FR-18	Copy, paste, cut, right-click, text selection and common shortcuts (Ctrl/Cmd+C/V/T/N/R, F12, etc.) are blocked.
FR-19	Live **webcam preview** is shown in a corner (no recording or upload, for privacy).
FR-20	A visible violation counter and toast warning appear for every violation; all violations are logged with timestamp and type.

3.5 Evaluation & Results

ID	Requirement
FR-21	Each answer is scored 0-10 by Gemini on relevance, correctness, depth and communication.
FR-22	Final score (out of 100) = average answer score x 10 minus 15 per violation, never below 0.

ID	Requirement
FR-23	Report shows: overall score with grade, penalty breakdown, per-question score, feedback and an ideal answer.
FR-24	Report shows overall strengths, areas to improve and a hiring-readiness verdict.
FR-25	Report can be downloaded as a PDF; candidate can start a new interview.

4. Non-Functional Requirements

Area	Requirement
UI / UX	Modern, premium design: dark theme with gradients and glassmorphism, smooth Framer Motion animations, clear typography, accessible contrast, responsive layout.
Usability	Setup in under one minute; every step has clear guidance; friendly error states (mic blocked, unsupported browser, AI rate limit).
Performance	Next question ready in about 2-4 s; Gemini Flash used for low latency.
Security	Gemini API key stays on the server (Next.js API routes, .env file); never exposed to the browser. Input is validated server-side.
Privacy	No login, no database: interview data lives only in the browser session. Webcam feed is never recorded or uploaded.
Compatibility	Full voice support in Chrome and Edge (desktop). Other browsers fall back to typed answers with a notice.
Reliability	Graceful retry on AI errors; fallback questions if Gemini is unavailable.

5. Technology Stack

Layer	Technology	Why
Framework	Next.js 15 (App Router) + React 19 + TypeScript	Frontend and secure backend API routes in one project; type safety.
Styling	Tailwind CSS v4	Fast, consistent, highly customisable design.
Animation	Framer Motion	Smooth transitions, animated avatar and voice visualiser.
Icons	Lucide React	Clean, consistent icon set.
AI	Google Gemini 2.5 Flash via @google/genai SDK	Free tier, fast, supports structured JSON output.
Speech-to-text	Web Speech API (SpeechRecognition)	Free, built into the browser, no extra keys.
Text-to-speech	Web Speech API (SpeechSynthesis)	Free, natural voices in Chrome / Edge.
Proctoring	Fullscreen API, Page Visibility API, window blur, MediaDevices (webcam)	Native browser APIs, no extra dependencies.
State	Zustand (persisted to sessionStorage)	Light-weight state across setup, interview and result pages.
PDF report	jsPDF	Client-side PDF generation of the results.

6. System Architecture & Flow

6.1 User flow

Landing → Setup (role, experience, mode) → System check (mic, camera, speaker) → Rules & consent → Full-screen interview (ask → listen → evaluate, repeated) → Results report → Download PDF / Retry.

6.2 Components

- **Browser (client):** UI pages, speech recognition and synthesis, proctoring monitor, webcam preview, session state, PDF export.
- **Next.js API routes (server):** */api/question* generates the next question; */api/evaluate* scores one answer; */api/report* produces the overall summary. All routes call Gemini with the server-side key and return structured JSON.
- **Google Gemini:** question generation and answer evaluation with JSON-schema constrained output.

6.3 Project structure

Path	Purpose
src/app/page.tsx	Landing page
src/app/setup/page.tsx	Setup wizard (role, experience, mode, system check, rules)
src/app/interview/page.tsx	Proctored voice interview room
src/app/result/page.tsx	Detailed results report
src/app/api/*	Gemini-backed API routes (question, evaluate, report)
src/hooks/*	useSpeechRecognition, useSpeechSynthesis, useProctoring
src/lib/*	Gemini client, prompts, scoring, PDF export
src/store/*	Interview session store

7. Scoring Model

Element	Rule
Per-question score	0-10, from Gemini (relevance, correctness, depth, communication)
Base score	Average of answered question scores x 10 (out of 100). Skipped questions count as 0.
Penalty	15 marks per violation (tab switch, window blur, full-screen exit)
Final score	max(0, Base score - 15 x violations)
Auto-termination	At 3 violations the interview ends immediately and is flagged.
Grade	85+ Excellent, 70-84 Good, 50-69 Average, below 50 Needs Improvement

8. Risks & Mitigations

Risk	Mitigation
Gemini free-tier rate limits when many users share one key	Use Flash model, compact prompts, friendly retry message, fallback questions.
Speech recognition unsupported (Firefox / Safari)	Detect support; show notice and allow typed answers.
False tab-switch detection (e.g. OS pop-ups)	Short debounce; clear warning toasts so candidates understand each penalty.
Browser proctoring can be bypassed (second device)	Accepted limitation for a practice tool; webcam preview increases realism.

9. Implementation Plan

Phase	Deliverable
1	Project scaffold, design system, landing page
2	Setup wizard and system check
3	Gemini API routes (question, evaluate, report)
4	Voice interview room (TTS, STT, avatar, timer)
5	Proctoring (full screen, tab switch, blocking, webcam, auto-end)
6	Results report and PDF download
7	Testing, polish and README with setup steps

9.1 Running the project

- Get a free API key at *aistudio.google.com/apikey*.
- Put it in *.env.local* as *GEMINI_API_KEY=...*
- Run *npm install* then *npm run dev* and open *http://localhost:3000* in Chrome.